



17-RESOURCE-BUDGETS.md — Budgets are assertions, not aspirations



Rule R4 — Budgets are assertions. Every number in this file is checked by code at runtime and by a job in CI. A performance target that lives only in a document is a wish; a target that fails the build is an engineering constraint.

1. Reference machines

Every budget is stated per tier. A claim without a tier is meaningless.

Tier	Spec	Role
T-Min	4-core CPU, 16 GB RAM, integrated GPU / 2 GB VRAM, SATA SSD	Must complete correctly. Slow is acceptable; failing is not.
T-Typ	8-core CPU, 32 GB RAM, 8 GB VRAM (RTX 4060 class), NVMe	The budgets below are enforced against this tier.
T-Max	16-core CPU, 64 GB RAM, 24 GB VRAM, NVMe RAID	Headroom validation and scaling checks.
T-Mac	Apple M-series, 32 GB unified memory	Unified-memory budgets differ; tracked separately.

2. Wall-clock budgets — 3,000-file wedding on T-Typ

Stage	Target	Hard ceiling (fails CI)
Import + hash + embedded previews	4 min	7 min
Tier-1 analysis (all brains)	6 min	9 min

Stage	Target	Hard ceiling (fails CI)
Cull decision + coverage guard	30 s	90 s
Tier-2 develop on 900 selects	12 min	18 min
Retouch + restoration on 900	18 min	28 min
Gallery consistency pass	3 min	5 min
QC agent + re-edit loop	5 min	9 min
Tier-3 full-res export (900 JPEG)	14 min	22 min
End-to-end autopilot	≤ 65 min	90 min

3. Interaction budgets (these are what users actually feel)

Action	p50	p95	Ceiling
App cold start to usable grid	1.2 s	2.5 s	4 s
Grid scroll frame time (3,000 thumbs)	8 ms	16 ms	16.7 ms — never drop below 60 fps
Open a photo to 1:1 preview	180 ms	400 ms	800 ms
Slider drag to updated preview	16 ms	33 ms	50 ms
Cancel a run	400 ms	2 s	3 s
UI thread blocked	0 ms	0 ms	50 ms — any longer is a bug

4. Memory ceilings

Resource	T-Min	T-Typ	Enforcement
Peak process RSS	3.5 GB	8 GB	Sampler asserts; run aborts to a checkpoint rather than being OOM-killed
Preview cache	512 MB	2 GB	Byte-budgeted LRU, never item-count

Resource	T-Min	T-Typ	Enforcement
Decode staging buffers	256 MB	1 GB	Pool with a fixed count; backpressure when exhausted
Peak VRAM	1.6 GB	6 GB	Tile solver + arena accounting
Inference sessions live	2	5	Derived from VRAM budget
Catalog size per 3,000 photos	≤ 220 MB	≤ 220 MB	Size assertion in the golden run
Disk cache per wedding	≤ 12 GB	≤ 12 GB	Pruned by age + budget; user-visible

Steady-state rule: RSS after a completed run must return to within 5 % of the pre-run baseline within 60 s. Anything else is a leak, and the leak test fails the build.

5. Cloud spend budget

Scope	Default cap	Behaviour at cap
Per 3,000-image wedding	75 calls / USD 1.50	Refuse further calls, switch to local fallback, mark Degraded
Per day	USD 10	Same, plus a UI notice
Per single call	8 s timeout, 1 retry ladder	Fallback
Images uploaded	≤ 40 proxies per wedding, max 1024 px	Hard block; counted in the run report

The governor is a **pre-flight check**, not a post-hoc audit: a call that would exceed the cap is never sent.

6. The budget assertion API (frozen)

```
// crates/aura-perf/src/budget.rs – FROZEN CONTRACT v1
pub struct Budget {
    pub name: &'static str,           // "stage.develop.p95_ms"
    pub tier: MachineTier,
    pub target: f64,
    pub ceiling: f64,                 // exceeding this fails CI
}
```

```
and logs AURA-JOB-7005
    pub unit: Unit,                // Ms | Bytes | Count | Usd
}

/// Records a sample and, in test/CI builds, fails if the ceiling is exceeded.
pub fn observe(budget: &'static Budget, value: f64);
```

All budgets live in one file, `perf/budgets.toml`, so the whole envelope can be read at a glance and diffed in review.

7. Benchmark discipline

- **Fixed hardware.** Benchmarks run on dedicated self-hosted runners, never on shared CI machines. A shared runner produces noise, and noise trains people to ignore the gate.
- **Warm-up + repeats.** Three warm-up runs discarded, then five measured; report median and p95, never a single run.
- **Thermal record.** Capture clocks and thermal state; a throttled run is annotated and excluded from regression comparison (F-20).
- **Regression gate.** A PR fails if median regresses **> 7 %** on any stage or **> 3 %** on interaction budgets versus the rolling 7-day baseline.
- **Improvements are also reviewed.** A sudden 40 % speed-up usually means work stopped happening. The PR must show output equality against the golden manifest.

8. Degradation ladder

When resources are short, AURA gets slower and honest, never broken and silent:

1. Reduce GPU tile size.
2. Reduce inference batch size.
3. Reduce live sessions.
4. Switch a stage from GPU to CPU (`Degraded`, logged).

5. Reduce preview cache to the floor.
6. Serialise stages entirely.
7. Pause with a clear message and a resume button — never crash, never silently skip photos.

The ladder is exercised by the resource-squeeze chaos scenario, and each rung must be individually reachable in tests.

9. Acceptance criteria

- ☐ `perf/budgets.toml` exists; every budget in this document is represented.
- ☐ `observe()` calls exist on every stage boundary and interaction path.
- ☐ Benchmark job runs on dedicated hardware with warm-up and repeats.
- ☐ Regression gate active at 7 % / 3 %.
- ☐ The full 3,000-file autopilot completes on T-Min without exceeding memory ceilings.
- ☐ Leak test: RSS returns to baseline $\pm$ 5 % after a run.
- ☐ Cloud governor blocks pre-flight and is proven by a test that attempts to exceed the cap.

10. Claude Code prompt

Act as `PERF` (Performance Engineer). Implement `crates/aura-perf` with the frozen `Budget` API and populate `perf/budgets.toml` from `17-RESOURCE-BUDGETS.md`. Instrument every stage boundary and the interaction paths. Build the benchmark harness with warm-up, repeats, thermal capture and the 7 % / 3 % regression gate. Then switch hats to `QAL` and implement the memory-ceiling and leak tests, plus a test that walks every rung of the degradation ladder. Report a table of budget name → measured → target → ceiling → PASS/FAIL. Never relax a ceiling to make a run pass; report the miss instead.